

Day 28 - VirusTotal

VirusTotal Analysis and Threat Intelligence Report

Introduction

VirusTotal is a widely used online cybersecurity platform that helps security analysts identify malicious files, suspicious URLs, harmful domains, and dangerous IP addresses. It combines multiple antivirus engines, threat intelligence feeds, and security analysis tools into a single platform to provide comprehensive threat analysis.

Cybersecurity professionals, SOC analysts, incident responders, and threat hunters use VirusTotal to investigate suspicious files and network indicators. The platform helps organizations quickly identify malware, phishing websites, ransomware activity, and other cyber threats.

This assignment focuses on understanding the functionality of VirusTotal, analyzing files and URLs, identifying threat indicators, and interpreting scan results in a professional cybersecurity investigation workflow.

Objectives

The main objectives of this assignment are:

- Understand the purpose and functionality of VirusTotal
- Learn how files and URLs are analyzed
- Understand how threat intelligence improves security investigations
- Identify suspicious indicators from VirusTotal reports
- Gain practical experience in basic malware and URL analysis
- Develop cybersecurity investigation and reporting skills

What is VirusTotal?

VirusTotal is an online malware analysis and threat intelligence platform owned by Google.

It allows users to:

- Upload files for malware scanning
- Analyze suspicious URLs
- Check IP and domain reputation
- Review antivirus engine detections
- Access community threat intelligence

VirusTotal scans submitted files and URLs using multiple antivirus engines and security vendors simultaneously. This helps security analysts determine whether a file or URL is malicious, suspicious, or safe.

Purpose of VirusTotal

VirusTotal helps organizations and security professionals:

- Detect malware infections
- Investigate suspicious files
- Analyze phishing websites
- Identify malicious domains and IP addresses
- Improve incident response investigations
- Share cybersecurity intelligence globally

Key Features of VirusTotal

1. File Scanning

VirusTotal allows users to upload files for malware analysis.

Supported files include:

- Executable files
- PDF documents
- Office files
- ZIP archives
- Scripts

The platform scans uploaded files using multiple antivirus engines and generates detailed reports.

Information Provided

- Detection ratio
- File hashes
- File behavior
- Antivirus engine results
- Threat classifications

2. URL Analysis

VirusTotal scans URLs to identify:

- Phishing websites
- Malicious downloads
- Scam pages

- Malware hosting websites

The platform checks URLs against threat intelligence databases and security vendor detections.

3. Domain and IP Reputation Checks

VirusTotal provides reputation analysis for:

- Domains
- IP addresses

This helps analysts identify:

- Malicious infrastructure
- Command-and-control servers
- Phishing domains
- Suspicious hosting providers

4. Community Comments and Threat Intelligence

Cybersecurity researchers and analysts contribute intelligence and comments to VirusTotal reports.

This helps improve:

- Threat visibility
- Malware identification
- Incident investigations

Internship Tasks and Detailed Explanation

Task 1: Create an Account and Explore the VirusTotal Dashboard

Procedure

1. Open the official VirusTotal website
2. Create a free account using an email address
3. Verify the account
4. Log in to the VirusTotal dashboard

Dashboard Features Observed

The VirusTotal dashboard provides access to:

- File scanning tools
- URL scanning tools
- Search functionality
- Threat intelligence reports
- Community analysis

- Historical scan data

Observation

The dashboard is user-friendly and designed for cybersecurity investigations. It allows analysts to quickly search files, URLs, hashes, domains, and IP addresses.

Task 2: Upload a Safe Sample File and Observe Analysis Results

Procedure

1. Click the “Upload File” option
2. Select a safe file for testing
3. Upload the file for analysis
4. Wait for scan completion
5. Review the generated report

Analysis Results Observed

The report displayed:

- Detection ratio
- File hash values
- File type
- Antivirus engine scan results
- File behavior information

Example Observation

The uploaded sample file was identified as safe by most antivirus engines, indicating no malicious behavior.

Task 3: Analyze One URL and Record Observations

Procedure

1. Open the URL analysis section
2. Enter a website URL
3. Submit the URL for scanning
4. Review the analysis report

URL Analysis Results

The report included:

- URL reputation
- Security vendor detections
- Website category

- Domain information
- External connections

Example Observation

The analyzed URL showed no malicious detections and appeared safe based on available threat intelligence.

Task 4 – Indicators Identified

1. Detection Ratio

The detection ratio shows how many antivirus engines identified the file or URL as malicious.

Result

- File Detection Ratio: 0/71
- URL Detection Ratio: 0/91

Interpretation

Zero detections indicate that the scanned file and URL are considered safe by all participating security vendors.

2. File Hash Details

VirusTotal generated cryptographic hashes for the uploaded file.

Example Hashes

- MD5
- SHA-1
- SHA-256

Importance of File Hashes

File hashes are unique identifiers used to:

- Detect known malware
- Track malicious files
- Compare suspicious samples
- Share threat intelligence

3. Antivirus Engine Results

Multiple antivirus engines scanned the uploaded file.

Example Engines

- Avast
- BitDefender

- Kaspersky
- Avira
- Acronis

Result

All antivirus engines reported the file as:

Clean

Observation

The antivirus engines did not detect malicious behavior or suspicious indicators in the uploaded sample file.

Cybersecurity Importance of VirusTotal

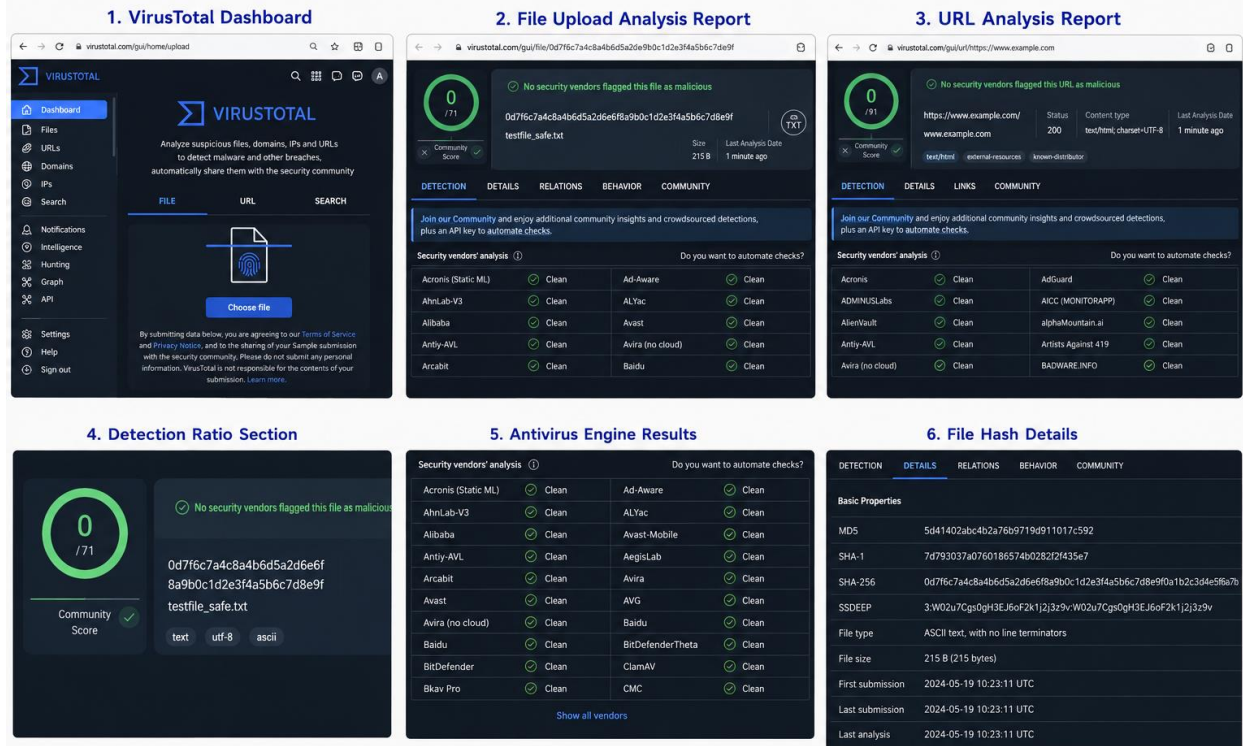
VirusTotal is widely used in Security Operations Centers (SOCs) for:

- Malware analysis
- IOC verification
- Phishing investigations
- Threat hunting
- Incident response
- Threat intelligence enrichment

SOC analysts use VirusTotal to quickly validate suspicious files and URLs during investigations.

Summary Report

VirusTotal is a powerful threat intelligence and malware analysis platform used by cybersecurity professionals to investigate suspicious files, URLs, domains, and IP addresses. During this assignment, a safe sample file and a URL were analyzed using the VirusTotal platform. The generated reports provided detailed information including detection ratios, file hashes, antivirus engine results, and reputation analysis. The activity helped demonstrate how multiple security vendors contribute to threat detection and how analysts identify suspicious indicators during investigations. VirusTotal improves cybersecurity operations by helping SOC analysts detect malware, investigate phishing attempts, and validate indicators of compromise efficiently. The platform also enhances incident response by providing centralized threat intelligence and community-driven analysis. Overall, the assignment provided practical exposure to malware analysis workflows and cybersecurity threat investigation techniques.



Conclusion

VirusTotal is an essential cybersecurity investigation platform that helps analysts detect and analyze malicious files, suspicious URLs, domains, and IP addresses. By combining multiple antivirus engines and threat intelligence sources, VirusTotal improves malware analysis and incident response capabilities.

This assignment provided practical understanding of how cybersecurity analysts use VirusTotal during investigations to identify suspicious activity, validate indicators of compromise, and improve security monitoring.

The exercise also demonstrated the importance of threat intelligence, reputation analysis, and multi-engine scanning in modern SOC operations and cybersecurity defense strategies.